

"Show Your Mind": Unveiling User Experience on an AI-based Mental Health Assessment System with Symptom-based Evidences

Hyunseon Won*
Sungkyunkwan University
Seoul, Republic of Korea
dprth1014@g.skku.edu

Daeun Lee
Sungkyunkwan University
Seoul, Republic of Korea
delee12@skku.edu

Daejin Choi
Computer Science and Engineering
Incheon National University
Incheon, Republic of Korea
djchoi@inu.ac.kr

Migyeong Kang*
Sungkyunkwan University
Seoul, Republic of Korea
gy77@g.skku.edu

Hyein Choi
Seongnam Citizens Medical Center
Seongnam, Republic of Korea
poonee82@gmail.com

Minsam Ko
Hanyang University ERICA
Ansan, Republic of Korea
minsam@hanyang.ac.kr

Minji Kim*
Sungkyunkwan University
Seoul, Republic of Korea
m5512m@g.skku.edu

Yonghoon Kim
OMNICNS
Seoul, Republic of Korea
yhhkim525@omnicns.com

Jinyoung Han[†]
Sungkyunkwan University
Seoul, Republic of Korea
jinyoung@skku.edu

Abstract

Online mental health assessment systems offer promise for individuals to evaluate their mental health without social stigma. With recent advancements, these systems evolved beyond pre-defined questionnaires to detect mental health conditions from user-generated text. However, existing research focused on model accuracy, with limited attention to user experiences. To bridge these gaps, we examine users' intention to adopt AI-based mental health assessment systems and investigate how symptom-based approaches affect user experience. We developed a mental health assessment system using natural language processing and conducted a within-subject study with 30 participants. Results demonstrated that symptom-based explanations enhance user's understanding of their mental health, with most participants expressing their intention to use. While accessibility, anonymity, and self-reflection positively influenced usage intention, the generalized result and lack of detailed explanation were a limiting factor. The findings suggest AI-based mental health assessment systems as supportive tools for early-stage evaluations, emphasizing the importance of personalized assessment.

CCS Concepts

• **Human-centered computing** → *User studies*; • **Computing methodologies** → *Information extraction*; • **Applied computing** → *Health care information systems*.

* Authors contributed equally to this research.

[†] corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '25, Yokohama, Japan
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1395-8/25/04
<https://doi.org/10.1145/3706599.3719735>

Keywords

Artificial Intelligence, Mental Health, Natural Language Processing, User Experience

ACM Reference Format:

Hyunseon Won, Migyeong Kang, Minji Kim, Daeun Lee, Hyein Choi, Yonghoon Kim, Daejin Choi, Minsam Ko, and Jinyoung Han. 2025. "Show Your Mind": Unveiling User Experience on an AI-based Mental Health Assessment System with Symptom-based Evidences. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3706599.3719735>

1 Introduction

Mental illness is a critical public health issue worldwide, as half of the global population is likely to experience a mental health problem at least once in their lifetime [61]. The urgency in this issue lies in the fact that many individuals needing treatment find it challenging to access appropriate care. In the United States, 90% of individuals who died by committing suicide had underlying mental health problems; however, many of them were either unaware of their conditions or did not receive adequate treatments [4]. Social stigma, defined as prejudice and discrimination against individuals with mental illnesses [31], has been identified as a major barrier to actively seeking treatment [53]. For instance, a survey conducted among university students reported that they often avoided mental health services despite having ample time [19], due to reluctance to reveal personal vulnerabilities [46] and fears of criticism for sharing negative emotions or failures [17, 57].

In response to these challenges, online mental health assessment systems have emerged as an innovative tool that allows individuals to assess their mental health without the stigma associated with seeking traditional treatment [52, 58]. These systems provide accessible and convenient environments where users can share their

symptoms more freely, contributing to early detection and treatment [34, 44, 62]. Traditionally, most online mental health assessment systems have relied on pre-defined questionnaires to collect user responses and generate aggregated scores to evaluate mental health conditions [1, 18, 20, 36]. With the rapid advancement in AI techniques, deep learning models that can analyze user-generated text in social media for mental health assessment have gained much attention [13, 41]. These models analyze natural language expressions, capturing nuanced contextual cues that structured questionnaires often overlook [8, 28, 48, 67]. Recent studies have introduced symptom-based approaches to enhance the explainability of these models [29, 38, 50, 70], which identify specific symptoms associated with mental health conditions within a user-generated text and provide them as evidence of predicted results. This methodology clarifies the reasoning behind assessments, significantly improving the explainability of mental health evaluation systems [5].

Despite technological advancements in AI-based mental health assessment, existing research has predominantly focused on improving model accuracy while giving less attention to the systematic exploration of user experience in such AI-based systems [23, 33]. Studies examining usability evaluations of online mental health systems have largely concentrated on conversational agents for providing emotional support to users [9, 22, 32], leaving significant gaps in research on AI-based mental health assessment. While a few studies have examined the usability of online mental health assessment platforms, these evaluations have mostly relied on the structured questionnaires [7, 37, 60] rather than analyzing individual status revealed in a free-text by AI techniques. Moreover, although symptom-based approaches have achieved high levels of explainability for mental health professionals [5, 21, 24, 27, 50], their effectiveness in enhancing users' comprehension and perceived utility of the assessment results remains unexplored. Consequently, understanding user experience on these AI-based mental health assessment systems remains underrepresented.

To bridge these gaps, we investigate user experiences in an AI-based mental health assessment system with symptom-based approaches. Specifically, the research aims to answer the two following questions: (i) whether providing symptom-based explanations enhances the user experience in mental health assessments, and (ii) what factors influence user's intention to adopt AI-based mental health assessment systems that analyze user-generated inputs. To this end, we developed an AI-based mental health assessment system that can provide symptom explanations as evidence of the evaluation results. We conducted a within-subject study with 30 adult participants interested in mental health to evaluate the system's effectiveness. The results demonstrated that symptom-based explanations significantly improved user experience, with transparency emerging as a key factor in enhancing user understanding. Furthermore, the majority of participants expressed their intention to use the system. Our qualitative analysis revealed accessibility, anonymity, and self-reflection derived from free-form text input are the significant factors positively influencing the user's intention to adopt online mental health assessment systems, whereas the generalized result and lack of detailed explanation, which may provide limited insights, was cited as a negative factor. These findings suggest the potential of AI-based mental health assessment systems show promise as a complementary tool in the early-stage mental

health screening process and underscore the need to develop more sophisticated and personalized assessment methods.

2 System Design

In this section, we introduce the process of developing an online mental health assessment system. We collected data from a mental health QA (Question and Answer) platform where users describe their mental state and seek expert opinions. The proposed framework leverages natural language processing techniques to identify signals for mental health conditions from user-generated input.

2.1 Dataset Construction

We constructed an annotated QA dataset for mental health assessment and symptom-based explanation using data from Naver Knowledge iN, a South Korean QA platform. We collected data from October 14, 2008, to July 15, 2021, focusing on three prevalent mental health conditions: Depression, Anxiety, and Eating disorders. We filtered the dataset to include only user-generated questions describing their mental state along with the diagnostic answers provided by experts. Three annotators – one with expertise in psychology and two in artificial intelligence – reviewed each user question-answer pair and annotated the dataset, assigning each user-generated question to a mental health condition diagnosed by an expert in their answer. Among different mental health conditions, we selected data assigned to three conditions that had sufficient samples to train a deep learning model for disorder detection: Depression, Anxiety, and Eating Disorders. We also maintained an “Others” category for cases unrelated to specific mental health conditions. The final dataset contained 2,335 QA pairs: depression (1,119), anxiety (649), eating disorders (567), and others (409). Personal information, such as names and email addresses, was anonymized to ensure privacy. Detailed dataset statistics and an example are available in the Appendix B.

2.2 Online Mental Health Assessment System

Using the constructed dataset, we developed an online mental health assessment system to assess user's mental health conditions and provide symptom-based explanations and coping strategies. The system consists of three main components: an assessment module, a symptom prediction module, and a coping strategy suggestion module, as illustrated in Figure 1.

2.2.1 Assessment Module. The assessment module is designed to detect signals of mental health conditions from user-generated text. To achieve this, we considered fine-tuning the KcBERT model [30], a specialized version of BERT [10] pre-trained on extensive Korean social media data, using the constructed dataset. However, given that KcBERT was primarily pre-trained on social media data, we first conducted an auxiliary task to supplement its lack of medical knowledge. We collected Korean medical information from Seoul National University Hospital (SNUH)¹ and Asan Medical Center (AMC)², resulting in 9,748 and 6,977 disease-related passage, respectively. Following the knowledge infusion methodology of He et al. [14], we generated question-answer templates in the format “What is the *[Aspect]* of *[Disease]*?” with masked tokens for diseases and

¹<http://www.snuh.org/health/nMedInfo/nList.do>

²<https://www.amc.seoul.kr/asan/healthinfo/disease/diseaseSubmain.do>

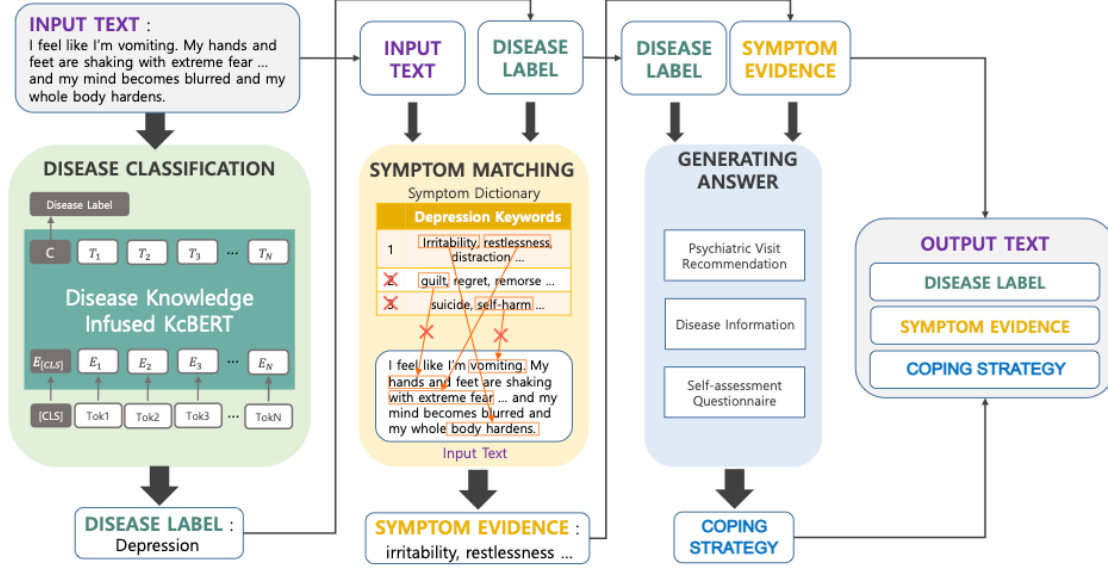


Figure 1: Components of the online mental health assessment system.

aspects, and trained KcBERT model to predict these masked tokens. After this auxiliary pre-training, we proceeded with the main fine-tuning to detect mental health conditions. This two-stage approach improved the model’s overall performance, increasing Accuracy by 1.82% (from 89.45% to 91.27%), F1-score by 1.82% (from 89.40% to 91.22%), and Recall by 1.82% (from 89.45% to 91.27%). Detailed results are provided in Table 1.

Model	Classes	Precision	Recall	F1-score
KcBERT	Depression	90.91%	87.38%	89.11%
	Anxiety	85.92%	91.04%	88.41%
	Eating Disorders	85.92%	98.39%	91.73%
	Others	100.00%	79.07%	88.31%
Accuracy		89.99%	89.45%	89.40%
KcBERT w/ Disease Knowledge Infusion	Depression	88.68%	91.26%	89.95%
	Anxiety	93.33%	83.58%	88.19%
	Eating Disorders	93.75%	96.77%	95.24%
	Others	91.11%	95.35%	93.18%
Accuracy		91.34%	91.27%	91.22%
Accuracy		91.27%		

Table 1: Performance results of the classification models for identifying mental health conditions.

2.2.2 Symptom Prediction Module. The symptom prediction module identifies symptom evidence from user-generated input using a symptom dictionary developed based on DSM-5 [2] diagnostic criteria. Although large language models have gained significant attention in various mental health tasks [63–65], their limitations

–including insufficient domain expertise and a tendency for over-diagnosis [65] – led us to instead adopt a reliable pattern matching approach using a symptom dictionary. To construct the symptom dictionary, we first extracted core symptom keywords for the three mental health conditions by referencing the DSM-5 (Diagnostic and Statistical Manual of Mental Disorders), which is used for diagnosing mental disorders in clinical settings. Then, we performed n-gram frequency analysis and Latent Dirichlet Allocation (LDA) topic modeling on our training dataset to identify patterns of how these symptoms are described in social media data. This approach enabled us to build a comprehensive symptom dictionary that not only covers the core symptoms outlined in DSM-5 for each condition but also incorporates how they are expressed by users in real-world contexts. After construction, the dictionary was validated using 100 labeled test samples per disorder, achieving an average accuracy of 85.08%. Detailed examples of the symptom dictionary and the validation process can be found in Appendix D.

2.2.3 Coping Strategy Suggestion Module. The coping strategy suggestion module provides personalized recommendations based on symptom severity calculated as the ratio of extracted symptoms to DSM-5 criteria. The system classifies severity into three levels: *Low risk* (<0.2), *Moderate risk* (0.2–0.5), and *Severe risk* (>0.5) with corresponding recommendations. Low-risk users are informed that a psychiatrist visit is recommended if needed, moderate-risk users are advised to seek psychiatric diagnosis, and severe-risk users are strongly encouraged to pursue mandatory psychiatric consultation and treatment. The module supplements these recommendations with additional information and standardized assessment tools for predicted mental health conditions, including the Center for Epidemiological Studies-Depression Scale (CES-D) [43], Beck Anxiety

Inventory (BAI) [3], and Eating Attitude Test (EAT-26) [12], enabling comprehensive self-evaluation and understanding.

3 Research Method

3.1 Study Design

We implemented the developed system as a web platform to enable direct user interaction and utilization. Figures 2a and 2b illustrate the system's user interface. The interface consists of an input screen where users can describe their mental health concerns (Figure 2a) and an output screen that displays the system's assessment results (Figure 2b). To examine how symptom-based explanations for mental health assessments affect user experience, we developed three system types based on the level of information provided: (1) assessment results only, (2) assessment results with symptom evidence, and (3) assessment results with symptom evidence and coping strategy suggestions.

The study followed a within-subject design. Participants interacted with all three system types in a randomized order to reduce learning effects and ordering bias. Each participant provided free-form input texts ranging from 200 to 600 characters for accurate assessment. After receiving answers from each system, participants completed an explainability survey. This process was repeated for all three system types. At the end of the study, participants answered user experience questionnaires and open-ended questions about their overall impressions of the system. Ethical approval for the study was obtained from the university Institutional Review Board (SKKU 2022-09-015), and all participants provided informed consent prior to participation.

3.2 Recruitment

We recruited participants for the study through a university website and social media platforms from August 11 to August 31, 2022. Eligibility criteria included being an adult with mental health concerns and having access to a laptop or computer with an internet connection. A total of 30 participants (23 females, 7 males) were recruited, with ages ranging from 20 to 53 years ($M=26.07$, $SD=6.12$). Educational backgrounds included 13 high school graduates, 16 college graduates, and one participant with a master's degree. All participants were provided with study guidelines and commenced the study after familiarizing themselves with the procedures. Upon completing all experiments and surveys, participants received \$7 USD as compensation.

3.3 Measurement

3.3.1 Explainability. Six factors were assessed to evaluate the system's explainability: Credibility, Trust, Transparency, Achievement, Usefulness, and Awareness. Credibility and Trust were measured using four and three items adapted from [11] and [54], respectively, with Cronbach's α values of 0.92 and 0.90. These factors reflect how the system's explanations influence users' confidence in the system and their trust in its results. Transparency, assessed with two items derived from [40] (Cronbach's $\alpha = 0.86$), examined whether the explanations helped users understand the rationale behind the system's assessment. Achievement and Perceived Usefulness were measured with two and three items based on [35] and [42], respectively, with Cronbach's α values of 0.87 and 0.95. These measures

assessed whether the system effectively addressed user concerns and provided valuable insights. Finally, Awareness was measured using two items adapted from [25] (Cronbach's $\alpha = 0.85$), evaluating whether the system's explanations increased users' understanding of their mental health. Participants completed these questionnaires after using each system, rating responses on a 7-point Likert scale ranging from 'strongly disagree' to 'strongly agree.'

3.3.2 Usability. Participants rated each system's usability using 7-point Likert scale questions on attitude, intention to use, ease of use, and satisfaction. Attitude and Intention to Use were measured with four and three items derived from [25] and [15], achieving Cronbach's α values of 0.96 and 0.95, respectively. Ease of Use and Satisfaction were assessed with four and two items from [35], with Cronbach's α values of 0.69 and 0.91, respectively.

3.3.3 Open-ended Questions. To investigate specific factors affecting usability, participants answered open-ended questions exploring five key areas: (i) overall impressions and evaluations of the system, including its strengths and weaknesses, (ii) preferred system type and the reasons behind their preference, (iii) how well the system's answers addressed the users' needs, (iv) the level of confidence in the system's diagnoses, and (v) suggestions for further system improvement.

4 Results

4.1 Quantitative Analysis

We conducted a statistical analysis of the questionnaire data using one-way repeated measures ANOVA (RM-ANOVA) to evaluate differences across the three system types, calculating *Eta-squared* to determine the magnitude of the effects. Post-hoc pairwise comparisons were performed using *Tukey's HSD*, with Cohen's *d* effect sizes calculated to provide additional context. The analyses were performed using *Pingouin* [56], an open-source statistical package for Python 3.

4.1.1 Effects of Explanation Levels. The results of RM-ANOVA (Figure 3) demonstrated that incorporating explanations in the system led to significant improvements across all six measured factors. The analysis revealed a particularly strong effect on Transparency, suggesting that explanations substantially enhanced users' understanding of the system's decision-making process. The progression from Type 2 (with symptom evidence) to Type 3 (with additional coping strategy information) yielded significant additional gains in Credibility, Achievement, and Awareness. However, for Trust, Transparency, and Usefulness, the addition of coping strategy in Type 3 did not produce statistically significant improvements beyond the gains already achieved with Type 2.

4.1.2 System Usability. We evaluated the usability of the proposed explainable online mental health assessment system based on user responses regarding Attitude, Intention to Use, Ease of Use, and Satisfaction. The survey results, summarized in Table 2, indicate an overall positive reception of the system. Attitude received a high average rating ($M = 4.62$, $SD = 1.71$), suggesting that participants enjoyed interacting with the system. Similarly, Intention to Use scored positively ($M = 4.66$, $SD = 1.85$), reflecting participants' willingness to continue using the system in the future. The system also

Please describe the situation you are currently experiencing

After AI analysis, we will suggest appropriate mental illness based on your input.

****Please write at least 200 characters and no more than 600 characters for accurate diagnosis****

Input Box

Currently 0 characters

Answer Button

(a) A free-text input box.

Assessment Result

This is the assessment result by artificial intelligence based on your situation.

hello:) It was a lot of trouble, but thank you for talking to me.

Mental Health Condition

Although it is difficult to be certain from your mentions, it is possible that you are experiencing depressive disorder.

Symptom Evidences

Complaining "worthlessness", "depression" is the main symptom of depressive disorders.

Coping Strategies

If the problem is so severe that it interferes with your daily life, we recommend that you visit the nearby Department of Psychiatric Medicine. You can also try the self-diagnosis test through the link below.

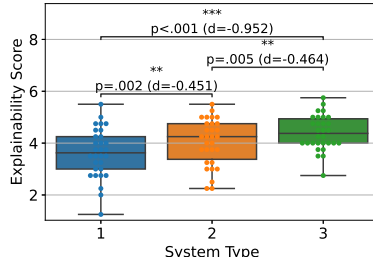
I hope our answer was helpful and lessened your worries

→ [Link to Depressive disorder disease information](#)

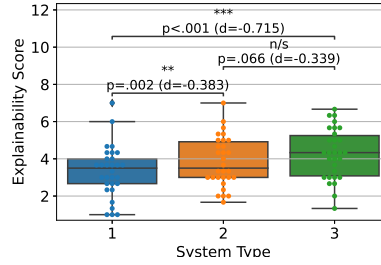
[Link to self-assessment of Depressive disorder](#)

(b) A sample answer provided by the system.

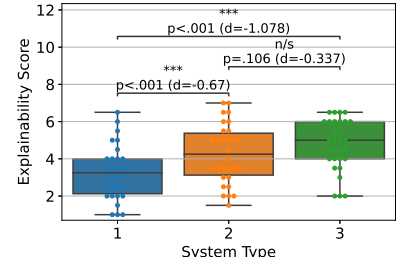
Figure 2: The system’s user interface, consisting of (a) a free-text input box and (b) a sample answer provided by the system.



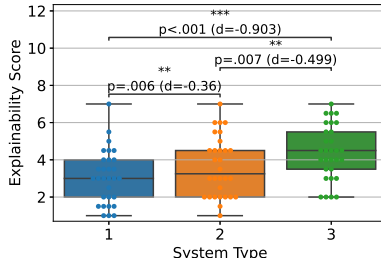
(a) Credibility ($p < .001$, $\eta^2 = 0.129$)



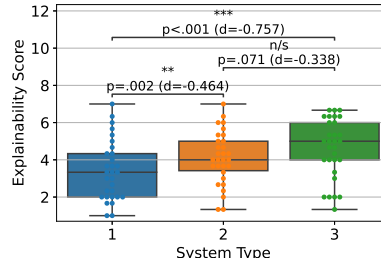
(b) Trust ($p < .001$, $\eta^2 = 0.082$)



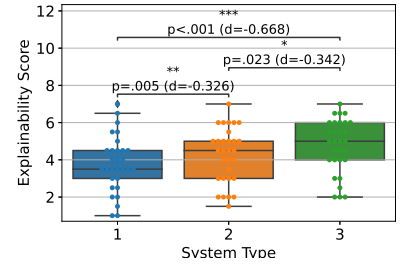
(c) Transparency ($p < .001$, $\eta^2 = 0.160$)



(d) Achievement ($p < .001$, $\eta^2 = 0.117$)



(e) Usefulness ($p < .001$, $\eta^2 = 0.097$)



(f) Awareness ($p < .001$, $\eta^2 = 0.071$)

Figure 3: Results of Repeated Measures ANOVA on Explainability.

demonstrated strong usability through high Ease of Use scores ($M = 5.53$, $SD = 1.35$), indicating its intuitive and user-friendly nature. Satisfaction ratings ($M = 4.75$, $SD = 1.71$) further confirmed users’ positive experience with the system. These results highlight the system’s effectiveness in providing a positive and accessible user experience.

4.2 Qualitative Analysis

To complement the quantitative findings, we analyzed the qualitative data from open-ended responses through thematic analysis. The responses identified the following five key components shaping user perceptions of the system.

4.2.1 Explainability. As the majority of participants preferred Type 3, the clarity was central to user satisfaction. For instance, P1 remarked, “The explanations helped me understand the connection between my recent symptoms and the diagnosis, which made me feel more comfortable.” The depth of the system’s explanations enhanced users’ ability to engage in meaningful self-reflection as P7 noted, “The detailed information on symptoms and the suggested assessments enabled a more thorough analysis, offering a valuable opportunity for self-reflection.”

4.2.2 Trust. Trust emerged as a recurring theme in user feedback, reflecting both confidence and skepticism. Many participants appreciated the anonymity of AI in diagnosing mental health conditions as participants felt more comfortable expressing their feelings without fear of judgment: *“It felt easier to be honest about my feelings when I was anonymously talking to an AI because I wasn’t speaking to a person.”* (P1). However, not all participants fully trusted the system. Some questioned AI’s capacity to replace human empathy, which they perceived as critical in mental health assessments *“AI could be helpful for initial assessments, but it cannot replace humans as mental health issues require empathy.”* (P11).

4.2.3 Awareness. Awareness refers to the system’s ability to help users reflect on their mental health and recognize potential issues. Many participants underscore the system’s ability to provide users with a structured opportunity for self-reflection. As P5 noted, *“Writing down my feelings freely helped me understand and myself more deeply.”* Similarly, P28 remarked, *“Thinking about the answers to the question helped me reflect on myself.”* However, the system’s generalized feedback occasionally hindered meaningful awareness. P15 commented, *“I think most people feel sleep-deprived and depressed, so I did not feel that the results were specific to my symptoms.”* These findings suggest that enhanced personalization could significantly improve its impact on users’ understanding of their mental health conditions.

4.2.4 Transparency. Transparency emerged as a critical component of user satisfaction, particularly in explaining how the system reached its conclusions. Participants emphasized the importance of symptom evidence that connected their input. P25 noted, *“It provides main keywords from my text and elaborates on the possible mental conditions”* However, some participants found the level of transparency inadequate, often citing a lack of depth in the explanations. P14 remarked, *“The reasons for the result were insufficient. Providing more detailed analysis on the free-text will make the result more reliable”* P9 shared, *“It felt like the system made a decision based on only a few keywords. The result was too simple.”* These critiques suggest that while the system provides a foundational level of transparency, more comprehensive and nuanced explanations are necessary to fully satisfy user expectations and foster trust in the assessment process.

4.2.5 Accessibility. Accessibility was a pivotal theme in evaluating the usability and effectiveness of the mental health assessment system. Participants frequently emphasized the convenience and inclusivity offered by the online platform. P27 remarked on overcoming barriers traditionally associated with seeking mental health support, *“In a society where visiting a psychiatrist can feel burdensome, this system makes it easier to take the first step toward understanding my mental health.”* Similarly, P12 shared a significant advantage of our system as the ability to access the platform from any location with an internet connection, stating *“It’s convenient to use this system for simple assessments without having to physically visit a clinic.”*

5 Discussion

5.1 User Experience in AI-based Mental Health Assessment

This study explored user experiences with AI-based mental health assessments, focusing on how symptom-based evidence improves

user experience and identifying the level of acceptance and the factors that influence users’ intentions to adopt the system. Our results demonstrated that providing symptom-based explanations significantly enhanced all measured aspects of explainability, with transparency showing the most notable improvement. The findings suggest that offering symptom evidence with the assessed mental health condition allows users to better interpret their mental health status and understand the system’s reasoning process. While the addition of behavioral suggestions and supplementary information showed significant improvements in credibility, achievement, and awareness, its effects on trust, transparency, and usefulness were less pronounced. These results imply that while solution suggestions enhance the depth of system feedback, their direct impact on core explainability factors may be more limited. Overall, the usability evaluation revealed high user satisfaction and strong intentions for future use, suggesting the potential of AI-based mental health assessment systems as accessible and supportive assistive tools that allow users to comfortably engage in mental health assessments.

Thematic analysis of qualitative responses provided additional insight into the factors influencing user perceptions and usage intentions. Participants appreciated the anonymity provided by the system, which reduced fears of judgment or stigma often associated with seeking mental health support, and accessibility, which enabled convenient use as an assistive tool without an in-person visit. These findings underscore the potential of an AI-based mental health assessment system to facilitate early detection of mental health conditions and bridge for individuals hesitant to seek in-person support. Furthermore, the use of free-form text inputs was highlighted as a unique strength, with participants describing the process of articulating their thoughts as introspective and therapeutic. Unlike traditional pre-defined questionnaire-based assessments, this approach encouraged users to explore their mental health more deeply, fostering self-reflection and personal insight. However, some participants expressed dissatisfaction with the system’s explanations, describing them as overly generic, which made it difficult for users to relate the assessment to their personal situations. These limitations highlight the need for further refinement of the system’s assessment models and explanation mechanisms to deliver more personalized feedback. Integrating large language models with advanced contextual understanding could address these concerns, mitigating the risk of over-diagnosis and enabling more individualized analyses of user inputs.

5.2 Limitations

Our study has several limitations. First, the system was limited to the assessment of three mental health conditions – depression, anxiety, and eating disorders. Second, the findings may have limited generalizability due to the small sample size and the specific characteristics of the participant pool. Additionally, as the study was not a clinical trial, it does not provide evidence of the system’s medical efficacy in diagnosing or treating mental health conditions. Future research should address these limitations through broader assessment coverage, larger and more diverse samples, and clinical validation to develop more comprehensive and reliable mental health assessment systems.

Factor	Item	Result		
		Mean (SD)	Mean (SD)	α
Attitude	“Using this system is a good idea.”	4.63 (1.76)	4.62 (1.71)	0.96
	“I have a generally favorable attitude toward using it.”	4.83 (1.55)		
	“Overall, using this system is beneficial.”	4.80 (1.60)		
	“I think a smartphone makes my life more interesting.”	4.20 (1.83)		
Intention to Use	“I will use it in future.”	4.80 (1.87)	4.66 (1.85)	0.95
	“I would like to recommend this program to others.”	4.70 (1.85)		
	“I will use it when I get a chance.”	4.47 (1.82)		
Ease of Use	“It is simple to use.”	5.73 (1.15)	5.53 (1.35)	0.69
	“It is easy to use.”	5.93 (0.99)		
	“It requires the fewest steps possible to accomplish my purpose.”	5.77 (1.02)		
	“I don’t notice any inconsistencies as I use it.”	4.70 (1.73)		
Satisfaction	“I am satisfied with it.”	4.77 (1.63)	4.75 (1.71)	0.91
	“It works the way I want it to work.”	4.73 (1.79)		

Table 2: Survey Responses on Usability of the Proposed System.

6 Conclusion

This paper investigated user experiences in an AI-based mental health assessment system that incorporates symptom-based explanations. Through experiments with 30 adult participants, we found that symptom-based explanations significantly improved users’ understanding and acceptance of the system. In particular, transparency emerged as a key factor in enhancing user experience, emphasizing the importance of explainability in designing AI-based mental health assessment systems. Our qualitative analysis also identified accessibility, anonymity, and opportunities for self-reflection through free-form text input as major positive factors influencing users’ intention to adopt the system. However, the generalized results limited the depth of insights, highlighting the need for more personalized assessment methods. The findings demonstrate the potential value of AI-based mental health assessment systems as an early-stage evaluation tool. Future work should involve systematic usability evaluations and clinical validation with broader user populations to ensure the safe and effective implementation of such systems.

Acknowledgments

This research was supported by the MSIT(Ministry of Science, ICT), Korea, under the Global Research Support Program in the Digital Field program(RS-2024-00425354) supervised by the IITP(Institute for Information & Communications Technology Planning & Evaluation) and the IITP-ITRC(Information Technology Research Center) grant funded by the Korea government(Ministry of Science and ICT)(IITP-2025-RS-2024-00436936).

References

- [1] Anish K Agarwal, Lauren Southwick, Arthur Pelullo, Haley J McCalpin, Rachel E Gonzales, David A Asch, Cecilia Livesey, Lisa Bellini, Rachel Kishton, Sarah Beck, et al. 2024. Cross-sectional analysis of healthcare worker mental health and utilisation of a digital mental health platform from 2020 to 2023. *BMJ Open Quality* 13, 3 (2024), e002896.
- [2] AP American Psychiatric Association, American Psychiatric Association, et al. 2013. Diagnostic and statistical manual of mental disorders: DSM-5.
- [3] Aaron T Beck, Norman Epstein, Gary Brown, and Robert A Steer. 1988. An inventory for measuring clinical anxiety: psychometric properties. *Journal of consulting and clinical psychology* 56, 6 (1988), 893.
- [4] Jonathan TO Cavanagh, Alan J Carson, Michael Sharpe, and Stephen M Lawrie. 2003. Psychological autopsy studies of suicide: a systematic review. *Psychological medicine* 33, 3 (2003), 395.
- [5] Siyuan Chen, Zhiling Zhang, Mengyue Wu, and Kenny Zhu. 2023. Detection of multiple mental disorders from social media with two-stream psychiatric experts.

In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. 9071–9084.

- [6] Arman Cohan, Bart Desmet, Andrew Yates, Luca Soldaini, Sean MacAvaney, and Nazli Goharian. 2018. SMHD: a large-scale resource for exploring online language usage for multiple mental health conditions. *arXiv preprint arXiv:1806.05258* (2018).
- [7] Jennifer Cronly, Alistair J Duff, Kristin A Riekert, Ivan J Perry, Anthony P Fitzgerald, Aine Horgan, Elaine Lehane, Barbara Howe, Muireann Ni Chroinin, and Eileen Savage. 2018. Online versus paper-based screening for depression and anxiety in adults with cystic fibrosis in Ireland: a cross-sectional exploratory study. *BMJ open* 8, 1 (2018), e019305.
- [8] Munmun De Choudhury, Michael Gamon, Scott Counts, and Eric Horvitz. 2013. Predicting depression via social media. In *Proceedings of the international AAAI conference on web and social media*, Vol. 7. 128–137.
- [9] Kerstin Denecke, Sayan Vaahesan, and Aaganya Arulnathan. 2020. A mental health chatbot for regulating emotions (SERMO)-concept and usability test. *IEEE Transactions on Emerging Topics in Computing* 9, 3 (2020), 1170–1182.
- [10] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. 4171–4186.
- [11] Mauricio S Featherman, Anthony D Miyazaki, and David E Sprott. 2010. Reducing online privacy risk to facilitate e-service adoption: the influence of perceived ease of use and corporate credibility. *Journal of services marketing* (2010).
- [12] David M Garner, Marion P Olmsted, Yvonne Bohr, and Paul E Garfinkel. 1982. The eating attitudes test: psychometric features and clinical correlates. *Psychological medicine* 12, 4 (1982), 871–878.
- [13] Sharath Chandra Guntuku, David B Yaden, Margaret L Kern, Lyle H Ungar, and Johannes C Eichstaedt. 2017. Detecting depression and mental illness on social media: an integrative review. *Current Opinion in Behavioral Sciences* 18 (2017), 43–49.
- [14] Yun He, Ziwei Zhu, Yin Zhang, Qin Chen, and James Caverlee. 2020. Infusing Disease Knowledge into BERT for Health Question Answering, Medical Inference and Disease Name Recognition. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 4604–4614.
- [15] Marcel Heerink, Ben Krose, Vanessa Evers, and Bob Wielinga. 2008. The influence of social presence on enjoyment and intention to use of a robot and screen agent by elderly users. In *RO-MAN 2008-The 17th IEEE International Symposium on Robot and Human Interactive Communication*. IEEE, 695–700.
- [16] Severin Hennemann, Sebastian Kuhn, Michael Withthöft, and Stefanie M Jungmann. 2022. Diagnostic performance of an app-based symptom checker in mental disorders: comparative study in psychotherapy outpatients. *JMIR Mental Health* 9, 1 (2022), e32832.
- [17] Janise A Hinson and Jane L Swanson. 1993. Willingness to seek help as a function of self-disclosure and problem severity. *Journal of Counseling & Development* 71, 4 (1993), 465–470.
- [18] Thomas D Hull, Matteo Malgaroli, Philippa S Connolly, Seth Feuerstein, and Naomi M Simon. 2020. Two-way messaging therapy for depression and anxiety: longitudinal response trajectories. *BMC psychiatry* 20 (2020), 1–12.
- [19] Justin Hunt and Daniel Eisenberg. 2010. Mental health problems and help-seeking behavior among college students. *Journal of adolescent health* 46, 1 (2010), 3–10.
- [20] Becky Inkster, Shubhankar Sarda, Vinod Subramanian, et al. 2018. An empathy-driven, conversational artificial intelligence agent (Wysa) for digital mental well-being: real-world data evaluation mixed-methods study. *JMIR mHealth and uHealth* 6, 11 (2018), e12106.

- [21] Hyolim Jeon, Dongje Yoo, Daeun Lee, Sejung Son, Seungbae Kim, and Jinyoung Han. 2024. A Dual-Prompting for Interpretable Mental Health Language Models. *arXiv preprint arXiv:2402.14854* (2024).
- [22] Eunkyung Jo, Daniel A Epstein, Hyunhoon Jung, and Young-Ho Kim. 2023. Understanding the benefits and challenges of deploying conversational AI leveraging large language models for public health intervention. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–16.
- [23] Juho Jung, Chaewon Kang, Jeewoo Yoon, Seungbae Kim, and Jinyoung Han. 2024. HiQuE: Hierarchical Question Embedding Network for Multimodal Depression Detection. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*. 1049–1059.
- [24] Migyeong Kang, Goun Choi, Hyolim Jeon, Ji Hyun An, Daejin Choi, and Jinyoung Han. 2024. CURE: Context-and Uncertainty-Aware Mental Disorder Detection. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. 17924–17940.
- [25] Ki Joon Kim and S Shyam Sundar. 2014. Does screen size matter for smartphones? Utilitarian and hedonic effects of screen size on smartphone adoption. *Cyberpsychology, Behavior, and Social Networking* 17, 7 (2014), 466–473.
- [26] Kurt Kroenke, Robert L Spitzer, and Janet BW Williams. 2001. The PHQ-9: validity of a brief depression severity measure. *Journal of general internal medicine* 16, 9 (2001), 606–613.
- [27] Daeun Lee, Hyolim Jeon, Sejung Son, Chaewon Park, Ji hyun An, Seungbae Kim, and Jinyoung Han. 2024. Detecting Bipolar Disorder from Misdiagnosed Major Depressive Disorder with Mood-Aware Multi-Task Learning. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. 4954–4970.
- [28] Daeun Lee, Migyeong Kang, Minji Kim, and Jinyoung Han. 2022. Detecting suicidality with a contextual graph neural network. In *Proceedings of the eighth workshop on computational linguistics and clinical psychology*. 116–125.
- [29] Daeun Lee, Sejung Son, Hyolim Jeon, Seungbae Kim, and Jinyoung Han. 2023. Towards Suicide Prevention from Bipolar Disorder with Temporal Symptom-Aware Multitask Learning. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 4357–4369.
- [30] Junbum Lee. 2020. KcBERT: Korean Comments BERT. In *Proceedings of the 32nd Annual Conference on Human and Cognitive Language Technology*. 437–440.
- [31] Inna Wanyin Lin, Lucille Njoo, Anjalie Field, Ashish Sharma, Katharina Reinecke, Tim Althoff, and Yulia Tsvetkov. 2022. Gendered mental health stigma in masked language models. *arXiv preprint arXiv:2210.15144* (2022).
- [32] Shuya Lin, Lingfeng Lin, Cuiqin Hou, Baijun Chen, Jianfeng Li, and Shiguang Ni. 2023. Empathy-based communication framework for chatbots: A mental health chatbot application and evaluation. In *Proceedings of the 11th International Conference on Human-Agent Interaction*. 264–272.
- [33] Yang S Liu, Jeffrey Hankey, Nigel Mantou Lou, Pratap Chokka, and Jason M Harley. 2021. Usability and emotions of mental health assessment tools: comparing mobile app and paper-and-pencil modalities. *Journal of Technology in Human Services* 39, 2 (2021), 193–211.
- [34] Gale M Lucas, Albert Rizzo, Jonathan Gratch, Stefan Scherer, Giota Stratou, Jill Boberg, and Louis-Philippe Morency. 2017. Reporting mental health symptoms: breaking down barriers to care with virtual human interviewers. *Frontiers in Robotics and AI* 4 (2017), 51.
- [35] Arnold M Lund. 2001. Measuring usability with the use questionnaire. *Usability interface* 8, 2 (2001), 3–6.
- [36] Nushka Marinova, Tim Rogers, and Angus MacBeth. 2022. Predictors of adolescent engagement and outcomes—A cross-sectional study using the togetherall (formerly Big White Wall) digital mental health platform. *Journal of affective disorders* 311 (2022), 284–293.
- [37] Robert G Maunder and Jonathan J Hunter. 2018. An internet resource for self-assessment of mental health and health behavior: development and implementation of the self-assessment kiosk. *JMIR mental health* 5, 2 (2018), e9768.
- [38] Thong Nguyen, Andrew Yates, Ayah Zirikly, Bart Desmet, and Arman Cohan. 2022. Improving the generalizability of depression detection by leveraging clinical questionnaires. *arXiv preprint arXiv:2204.10432* (2022).
- [39] Sungjoon Park, Kiwoong Park, Jaimeen Ahn, and Alice Oh. 2020. Suicidal risk detection for military personnel. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2523–2531.
- [40] Pearl Pu, Li Chen, and Rong Hu. 2011. A user-centric evaluation framework for recommender systems. In *Proceedings of the fifth ACM conference on Recommender systems*. 157–164.
- [41] Jiang Qiao. 2020. A systematic review of machine learning approaches for mental disorder prediction on social media. In *2020 International Conference on Computing and Data Science (CDS)*. IEEE, 433–438.
- [42] Emilee Rader, Kelley Cotter, and Janghee Cho. 2018. Explanations as mechanisms for supporting algorithmic transparency. In *Proceedings of the 2018 CHI conference on human factors in computing systems*. 1–13.
- [43] Lenore Sawyer Radloff. 1977. The CES-D scale: A self-report depression scale for research in the general population. *Applied psychological measurement* 1, 3 (1977), 385–401.
- [44] Abhilasha Ravichander and Alan W Black. 2018. An empirical study of self-disclosure in spoken dialogue systems. In *Proceedings of the 19th annual SIGDial meeting on discourse and dialogue*. 253–263.
- [45] Bahae Samhan. 2019. Self-Diagnosis Mobile Applications A Technology Trust Perspective. (2019).
- [46] Charles W Schmidt. 2007. Environmental connections: a deeper look into mental illness.
- [47] Ivan Sekulić and Michael Strube. 2020. Adapting deep learning methods for mental health prediction on social media. *arXiv preprint arXiv:2003.07634* (2020).
- [48] Judy Hanwen Shen and Frank Rudzicz. 2017. Detecting anxiety through reddit. In *Proceedings of the Fourth Workshop on Computational Linguistics and Clinical Psychology — From Linguistic Signal to Clinical Reality* (Vancouver, BC). 58–65.
- [49] Chaitali Sinha, Saha Meheli, Madhura Kadaba, et al. 2023. Understanding digital mental health needs and usage with an artificial intelligence-led mental health app (Wysa) during the COVID-19 pandemic: Retrospective analysis. *JMIR Formative Research* 7, 1 (2023), e41913.
- [50] Hoyun Song, Jisu Shin, Huije Lee, and Jong C Park. 2023. A simple and flexible modeling for mental disorder detection by learning from clinical questionnaires. *arXiv preprint arXiv:2306.02955* (2023).
- [51] Robert L Spitzer, Kurt Kroenke, Janet BW Williams, and Bernd Löwe. 2006. A brief measure for assessing generalized anxiety disorder: the GAD-7. *Archives of internal medicine* 166, 10 (2006), 1092–1097.
- [52] K Subramaniam, A Greenshaw, and A Thapliyal. 2024. The opportunity for e-mental health to overcome stigma and discrimination. *European Psychiatry* 67, S1 (2024), S549–S550.
- [53] Michael M Tadesse, Hongfei Lin, Bo Xu, and Liang Yang. 2019. Detection of depression-related posts in reddit social media forum. *IEEE Access* 7 (2019), 44883–44893.
- [54] Chun-Hua Tsai and Peter Brusilovsky. 2021. The effects of controllability and explainability in a social recommender system. *User Modeling and User-Adapted Interaction* 31, 3 (2021), 591–627.
- [55] Chun-Hua Tsai, Yue You, Xinning Gui, Yubo Kou, and John M Carroll. 2021. Exploring and promoting diagnostic transparency and explainability in online symptom checkers. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–17.
- [56] Raphael Vallat. 2018. Pingouin: statistics in Python. *J. Open Source Softw.* 3, 31 (2018), 1026.
- [57] David L Vogel and Stephen R Wester. 2003. To seek help or not to seek help: The risks of self-disclosure. *Journal of counseling psychology* 50, 3 (2003), 351.
- [58] Emma Wallin, Pernilla Maathz, Thomas Parling, and Timo Hursti. 2018. Self-stigma and the intention to seek psychological help online compared to face-to-face. *Journal of clinical psychology* 74, 7 (2018), 1207–1218.
- [59] Akash R Wasil, Katherine E Venturo-Conerly, Rebecca M Shingleton, and John R Weisz. 2019. A review of popular smartphone apps for depression and anxiety: assessing the inclusion of evidence-based content. *Behaviour research and therapy* 123 (2019), 103498.
- [60] Arne Weigold, Ingrid K Weigold, Naomi M Drakeford, Stephanie A Dykema, and Charity A Smith. 2016. Equivalence of paper-and-pencil and computerized self-report surveys in older adults. *Computers in Human Behavior* 54 (2016), 407–413.
- [61] Harvey A Whiteford, Alize J Ferrari, Louisa Degenhardt, Valery Feigin, and Theo Vos. 2015. The global burden of mental, neurological and substance use disorders: an analysis from the Global Burden of Disease Study 2010. *PLoS one* 10, 2 (2015), e0116820.
- [62] Kieran Woodward, Eiman Kanjo, David Brown, Thomas M McGinnity, Becky Inkster, Donald MacIntyre, and Thanasis Tsanas. 2020. Beyond mobile apps: a survey of technologies for mental well-being. *IEEE Transactions on Affective Computing* (2020).
- [63] Xuhai Xu, Bingsheng Yao, Yuanzhe Dong, Saadia Gabriel, Hong Yu, James Hendler, Marzyeh Ghassemi, Anind K Dey, and Dakuo Wang. 2024. Mental-llm: Leveraging large language models for mental health prediction via online text data. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 8, 1 (2024), 1–32.
- [64] Kailai Yang, Shaoxiong Ji, Tianlin Zhang, Qianqian Xie, Ziyang Kuang, and Sophia Ananiadou. 2023. Towards interpretable mental health analysis with large language models. In *The 2023 Conference on Empirical Methods in Natural Language Processing*.
- [65] Kailai Yang, Tianlin Zhang, Ziyang Kuang, Qianqian Xie, Jimin Huang, and Sophia Ananiadou. 2024. MentalLaMA: Interpretable Mental Health Analysis on Social Media with Large Language Models. In *Proceedings of the ACM on Web Conference 2024*. 4489–4500.
- [66] Zichao Yang, Diyi Yang, Chris Dyer, Xiaodong He, Alex Smola, and Eduard Hovy. 2016. Hierarchical attention networks for document classification. In *Proceedings of the 2016 conference of the North American chapter of the association for computational linguistics: human language technologies*. 1480–1489.
- [67] Andrew Yates, Arman Cohan, and Nazli Goharian. 2017. Depression and self-harm risk assessment in online forums. *arXiv preprint arXiv:1709.01848* (2017).

- [68] Yue You and Xinning Gui. 2021. Self-diagnosis through AI-enabled chatbot-based symptom checkers: user experiences and design considerations. In *AMIA Annual Symposium Proceedings*, Vol. 2020. 1354.
- [69] Jordyn Young, Laala M Jawara, Diep N Nguyen, Brian Daly, Jina Huh-Yoo, and Afsaneh Razi. 2024. The Role of AI in Peer Support for Young People: A Study of Preferences for Human-and AI-Generated Responses. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–18.
- [70] Zhiling Zhang, Siyuan Chen, Mengyue Wu, and Kenny Q Zhu. 2022. Symptom identification for interpretable detection of multiple mental disorders. *arXiv preprint arXiv:2205.11308* (2022).

A Related Work

A.1 Mental Health System with Questionnaire-based Assessment

With the growing prevalence of mental health issues, online mental health assessment systems have emerged to provide self-screening without the need for in-person consultations. According to Wasil et al. [59], 37% of surveyed online mental health services include self-assessment tools, typically relying on pre-designed questionnaires. Platforms such as Talkspace utilize the Patient Health Questionnaire (PHQ-9) [26] for depression, and the Generalized Anxiety Disorder Scale (GAD-7) [51] for anxiety as part of its two-way messaging therapy [18]. Similarly, Wysa [20, 49] and Togetherall [36] incorporate PHQ-9 and GAD-7 for mental health assessments. Cobalt, a mental health platform at the University of Pennsylvania Health System, evaluates conditions including depression, anxiety, and PTSD using similar tools [1]. While these standardized questionnaires facilitate rapid and scalable assessments, their reliance on fixed questions limits the ability to capture nuanced, contextual insights from user inputs, often oversimplifying mental health evaluations. To overcome these limitations, researchers have turned to natural language processing (NLP) approaches that analyze user-generated data, offering richer, context-aware insights by identifying subtle linguistic cues.

A.2 Mental Health Assessment using AI Techniques

The increasing use of social media to share mental health experiences has spurred significant research into detecting mental health conditions from user-generated data. Early studies used platforms like Twitter [8] and Reddit [67] to classify depression and self-harm risk, employing models such as support vector machines and convolutional neural networks (CNNs). Cohan et al. [6] developed a SMHD dataset for nine distinct mental disorders, and Sekulić and Strube [47] showed that hierarchical attention networks (HAN) [66] were effective for classifying multiple mental disorders in the dataset. Furthermore, studies have also leveraged models such as BERT [10] to predict suicide risk among military personnel [39]. Recent advancements have focused on symptom-based methodologies that integrate standardized tools to enhance the explainability of deep learning models. Nguyen et al. [38] incorporated PHQ-9 symptoms into their model, while Zhang et al. [70] developed the Psysym dataset, mapping symptoms to seven mental disorders based on DSM-5 [2]. Song et al. [50] proposed a flexible framework using Siamese networks to extend symptom-based methods across multiple conditions. Kang et al. [24] demonstrated enhanced precision when enriched with contextual information such as the duration

and frequency of symptoms. However, despite their technical accuracy, such models often lack user-centered evaluations, particularly regarding user trust and acceptance.

A.3 User-Centered Studies in Mental Health Systems

With the expansion of online mental health systems, usability research in this domain has gained attention, primarily focusing on the effectiveness of conversational systems for emotional support. For example, SERMO was developed to help users regulate emotions [9], while empathy-based chatbots improved user satisfaction by incorporating emotion recognition and empathy-generation modules [32]. Systems like CareCall reduced loneliness among socially isolated populations [22], and studies on adolescent users found AI responses were often preferred over human responses for less sensitive topics [69]. Some studies have analyzed self-assessment systems called online symptom checkers, but these have predominantly focused on physical symptoms, with limited attention to mental health [45, 55, 68]. Existing research on mental health systems often emphasizes diagnostic accuracy, neglecting how users interpret and accept the system results [16]. This study aims to fill these gaps by developing an explainable mental health assessment system and systematically evaluating how different levels of explanation affect user understanding and acceptance. In addition, it analyzes users’ usage intentions and their influencing factors within the context of NLP-based mental health systems. The findings contribute to establishing design principles for mental health systems and provide empirical evidence for creating user-centered, explainable AI systems.

B Details of Data Construction

B.1 Data Statistics

We curated a dataset of mental health-related question-answer (QA) pairs from Naver Knowledge iN, covering the period from October 2008 to July 2021. The dataset comprises 2,744 QA pairs, focusing on three major mental health conditions: depression (1,119 pairs), anxiety (649 pairs), and eating disorders (567 pairs). Additionally, 409 entries were labeled as “Others,” representing cases that did not align with these three conditions. For model development and evaluation, the dataset was divided into training (70%), validation (20%), and test (10%) sets. As shown in Table 3, the training set contains 1,920 QA pairs, while the validation and test sets contain 549 and 275 pairs respectively.

Split	Total	Depression	Anxiety	Eating	Others
Train	1,920	787	455	404	274
Valid	549	229	127	101	92
Test	275	103	67	62	43
Total	2,744	1,119	649	567	409

Table 3: A summary of the collected dataset.

B.2 Data Example

Our dataset consists of user-generated questions in which individuals describe their mental health conditions, paired with corresponding expert answers. These expert responses include diagnostic

assessments, which served as the foundation for our annotations. To ensure labeling accuracy, the psychiatrist conducted additional reviews for cases where the assessments were ambiguous. Table 4 shows an example from our collected dataset.

C Details of Assessment Module

C.1 Example of Knowledge Infusion Training

We enhanced the KcBERT model’s medical domain knowledge through an auxiliary pre-training task using medical information collected from SNUH (9,748 passages) and AMC (6,977 passages). Following the approach of He et al. [14], we transformed these medical passages into masked language modeling (MLM) training samples. Table 5 illustrates the process of converting medical passages into training instances. For each passage, we identified the disease name and its corresponding aspect (e.g., definition, symptoms, treatment). We then reformulated the passage by prepending a question template “What is the [Mask] of [Mask]?”, where the masks correspond to the aspect and disease, respectively. This approach enables the model to learn the relationship between medical concepts and their associated aspects.

D Details of Symptom Prediction Module

D.1 Symptom Dictionary

Table 6 illustrates the structure and content of the symptom dictionary developed for identifying depression-related symptoms. This

dictionary is based on the DSM-5 diagnostic criteria and was refined through expert consultation, latent Dirichlet allocation (LDA), and n-gram analysis to ensure comprehensiveness and relevance. The table outlines specific DSM-5 criteria for depression alongside corresponding symptom keywords and user expression examples.

D.2 Validation Results of Symptom Prediction

The performance of the symptom extraction algorithm was validated using manually annotated test data. For this purpose, 100 test instances were randomly selected from each label category, and three annotators labeled symptom keywords within the input text. Verification and corrections were provided by a psychiatrist. This annotation process ensured that one or more symptom keywords were associated with each test instance. The algorithm’s outputs were evaluated for consistency with the annotated keywords. As a result, this module showed reasonable performance for each disease (depression disorder: P-92.30% & R-87.45%, anxiety disorder: P-83.25% & R-72.26%, eating disorder: P-79.66% & R-77.34%; P=Precision, R=Recall), showing average 85.08% precision and 79.02% recall.

Data Example	
User Question	It's concerning and distressing that whenever I go to crowded places, I experience panic attacks that make it impossible for me to leave my home. Ironically, while I genuinely enjoy conversing with people and meeting them face-to-face, these recurring attacks not only inconvenience the people I meet but have also led me to develop social anxiety as I fear what might happen. I'm wondering if these symptoms might indicate panic disorder?
Expert Answer	Hello. I'm Dr. Kim, a psychiatrist with the Korean Medical Association. When someone describes experiencing panic attacks in crowded places, palpitations, shortness of breath, and distressing symptoms, these may indicate symptoms of panic disorder. The fact that these symptoms occur specifically in crowded places could suggest agoraphobia. [...] For the treatment of panic disorder and anxiety, options include cognitive behavioral therapy and medication, among other approaches. I recommend consulting a mental health professional for proper evaluation, counseling, and treatment.

Table 4: An example of a question and an answer.

Type	Context
Original Passage	Anxiety disorder is a collective term for mental disorders that cause disturbances in daily life due to various types of abnormal and pathological anxiety and fear. Anxiety and fear are normal emotional reactions as warning signs ...
Disease	Anxiety
Aspect	Definition
New Passage for MLM	What is the [Mask] of [Mask]? Anxiety disorder is a collective term for mental disorders that cause disturbances in daily life due to various types of abnormal and pathological anxiety and fear. Anxiety and fear are normal ...

Table 5: An Example of Constructing a Knowledge Infusion Training Resource.

DSM-5 Criteria	Symptom Keywords and User Expression Examples
Depressed mood most of the day, nearly every day	keywords: tears, melancholy, despair, emptiness, emptiness “I often feel depressed”
Markedly diminished interest or pleasure in all, or almost all, activities most of the day, nearly every day.	keyword: boring “I’m tired of not having fun even playing”
Significant weight loss when not dieting or weight gain, or decrease or increase in appetite nearly every day.	keywords: appetite, weight, gluttony, taste “I binge eat once I start”
Insomnia or hypersomnia, nearly every day	keywords: insomnia, sleep “I can’t sleep well if I don’t take medicine”
A slowing down of thought and a reduction of physical movement	keywords: irritability, delay, restlessness, distraction “I’m so nervous all day I can’t stay still”
Fatigue or loss of energy nearly every day.	keywords: lethargy, annoyance, tiredness, fatigue “I’m lethargic all day and I’m not motivated to do anything”
Feelings of worthlessness or excessive or inappropriate guilt nearly every day.	keywords: meaninglessness, skepticism, guilt, pity, regret, remorse “I feel life is meaningless”
Diminished ability to think or concentrate, or indecisiveness, nearly every day.	keywords: dyslexia, forgetfulness, indecisiveness “I have trouble concentrating and keep having depressing thoughts”
Recurrent thoughts of death, or a suicide attempt or a specific plan for committing suicide.	keywords: death, life, suicide, self-harm “Nothing to live for”

Table 6: Symptom Keywords and Expression Example of Depression.